



Society of Petroleum Engineers

**SPE-229995-MS**

## **Using Hydrocarbon and Water Isotopes Analysis to Complement Production Logging Tools: A Case Study from the X Field, Oman**

Sheikha Abdullah Suhail Al Maamari and Ramzi Ibrahim Abdulkadir, Shell Development Oman LLC, Muscat, Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229995-MS](https://doi.org/10.2118/229995-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

---

### **Abstract**

Accurate reservoir characterization and fluid allocation are critical challenges in hydrocarbon production, especially in fields with complex geology and commingled reservoirs. Traditionally, Production Logging Tools (PLTs) have been the primary diagnostic method for determining zonal contributions, but these tools are often limited by operational constraints, high costs, and the need for well interventions. In recent years, isotopic fingerprinting of hydrocarbons and formation water has emerged as a complementary method that can enhance the understanding of reservoir performance with minimal intrusion.

This study investigates the integration of hydrocarbon gas and water isotope fingerprinting with conventional PLT data in the X Field, Oman—a field characterized by stacked, commingled production from the Reservoir A and Reservoir B formations. Surface samples collected between 2022 and 2024 from selected wells across the field were analyzed for chemical and isotopic signatures, including  $\delta^{13}\text{C}$  in methane (C1) and ethane (C2), as well as strontium isotope ratios in produced water. These isotopic markers were used to establish geochemical endmembers and mixing lines that distinguish between Reservoir A and Reservoir B fluid sources.

The results indicate that while gas isotope analysis provided limited resolution due to overlapping  $\delta^{13}\text{C}$  values, the integration of strontium isotopes in water and compositional fingerprinting of condensates enabled clear differentiation between reservoir contributions. Several wells showed isotopic profiles that confirmed or clarified ambiguous PLT interpretations, improving confidence in zonal allocation. The integration of these datasets reduced the need for additional PLT runs and enabled more cost-effective reservoir surveillance.

This study demonstrates the practical application and added value of isotopic fingerprinting in mature fields. It offers a scalable, low-impact methodology for improving production allocation and optimizing field development strategies in complex reservoir systems.

### **Introduction**

In the upstream oil and gas industry, maximizing recovery while minimizing operational costs relies heavily on accurate reservoir characterization and production allocation. Understanding which reservoir zones are contributing hydrocarbons—and in what proportions—is essential for making informed decisions around

reservoir management, well interventions, and field development planning. Conventional techniques such as Production Logging Tools (PLTs) have long been used to monitor production profiles, evaluate inflow zones, and diagnose well performance issues. However, PLTs are not without their challenges. These tools often require costly well interventions, provide only snapshot data during specific well conditions, and may be limited in resolution—particularly in highly deviated wells or in fields with closely spaced stacked reservoirs and overall relatively small water (BS&W).

In recent years, advances in geochemical and isotopic analysis have opened new avenues for reservoir diagnostics. By analyzing the unique chemical and isotopic signatures of produced fluids, operators can infer the origin of hydrocarbons and formation water, detect compartmentalization, and quantify commingled flow contributions. Isotopic fingerprinting, in particular, has proven valuable in distinguishing fluids from different formations or reservoir compartments. Isotopes such as  $\delta^{13}\text{C}$  in methane and ethane, and strontium isotopes in formation water, provide robust tracers that are not significantly altered by surface processing and can therefore serve as reliable indicators of subsurface processes and reservoir performance.

This paper presents a case study from the Field in Oman, where isotope fingerprinting was integrated with PLT data to enhance production allocation and reservoir understanding. The Field consists of multiple stacked gas-condensate reservoirs, including the Reservoir A and Reservoir B formations, which produce commingled through shared completions. Historical reliance on PLTs in this field has provided useful inflow profiles, but challenges such as high water cut, reservoir compartmentalization, and limited PLT resolution have prompted the search for alternative or supplementary surveillance methods.

Over a two-year period (2022–2024), a systematic fluid sampling campaign was conducted across multiple wells, collecting produced gas, condensate, and formation water samples. These were analyzed for stable carbon isotopes in hydrocarbon gases and for strontium isotopes and ionic composition in produced water. The objective was to determine whether isotopic and geochemical signatures could be used to identify the source formation of produced fluids, validate or refine PLT-based interpretations, and ultimately reduce the frequency and cost of PLT operations.

The results from this study demonstrate that isotopic fingerprinting provides a technically robust and economically efficient complement to PLT data. The use of strontium isotopes, in particular, enabled the identification of distinct water families corresponding to the Reservoir A and Reservoir B formations, and helped clarify zonal contributions in wells where PLT data were ambiguous. This integrated approach not only improves the confidence in production allocation but also supports more informed field development strategies.

## Field Background

The Field is located in central Oman and forms part of a mature hydrocarbon province that has been producing gas and condensate for several decades. The field is geologically complex, comprising multiple stacked sandstone reservoirs that include Reservoir A and Reservoir B formations; both members of the Haima Supergroup. These formations were deposited during the late Cambrian to early Ordovician period and are characterized by moderate-to-high porosity sandstones interbedded with shale and siltstone layers.

Reservoir A and Reservoir B reservoirs are typically encountered at depths ranging from 4,500 to 5,200 meters true vertical depth (TVD), with Reservoir B lying stratigraphically below Reservoir A. These formations exhibit good reservoir quality, although Reservoir B tends to have slightly lower permeability due to finer grain size and a higher degree of diagenetic cementation. Both formations are overpressured and thermally mature, producing dry gas and gas condensate with minimal oil.

One of the operational complexities in the Field arises from the commingled production strategy adopted in many wells. To maximize productivity and minimize the need for multiple completions, several wells have been completed across both Reservoir A and Reservoir B formations using shared tubing and perforation intervals. While this approach improves efficiency in field development, it creates challenges for

zonal allocation and reservoir management, as it becomes difficult to attribute produced fluids to a specific formation without intrusive diagnostics.

Historically, Production Logging Tools (PLTs) have been used periodically to determine zonal inflow contributions in selected wells. These PLTs are typically carried during shut-in or flow passes and provide a snapshot of the flow profile across the perforated interval. However, due to high costs, operational risks, and the inability to continuously monitor changing reservoir conditions, PLTs are used sparingly. Moreover, the resolution of PLTs may not always be sufficient to distinguish between closely spaced flow zones, especially in highly deviated or horizontal wells.

In recent years, the field has experienced an increasing trend in water production, with some wells exhibiting high water cuts that complicate reservoir performance interpretation. This has further underscored the need for alternative surveillance techniques that can complement PLTs, reduce intervention frequency, and enhance reservoir insight.

To address these challenges, the field operator initiated a geochemical and isotopic fingerprinting program aimed at using formation water and hydrocarbon fluid signatures to trace the origin of production. By identifying isotopic endmembers associated with the Reservoir A and Reservoir B formations, the goal was to develop a non-intrusive, cost-effective method to augment traditional surveillance data and support more accurate production allocation.

## Methodology

To evaluate the effectiveness of isotopic fingerprinting as a complementary tool for reservoir surveillance in the Field, a structured multi-phase methodology was implemented. The process involved field-wide sampling of hydrocarbons and formation water, laboratory analysis for isotopic and chemical composition, development of isotopic endmember models, and integration of these results with existing PLT data.

### Fluid Sampling Campaign

A comprehensive fluid sampling campaign was carried out between 2022 and 2024, targeting a representative spread of wells across the Field. The selection of wells was based on geographic distribution, production performance, reservoir completion history, and PLT availability. A total of 24 wells were sampled, with specific focus on those completed in both Reservoir A and Reservoir B formations.

From each well, three types of samples were collected:

- **Gas samples**, captured at the separator using gas sampling cylinders.
- **Condensate samples**, collected from liquid phase separators under stabilized flow conditions.
- **Produced water samples**, collected from test separators, taking care to minimize contamination from surface processing or tank residuals.

All samples were labeled, preserved, and shipped under controlled conditions to certified geochemistry laboratories for analysis.

### Gas Isotope Analysis and Gas Compositional Analysis

Gas samples were analyzed for stable carbon isotopes of methane (C1) and ethane (C2) and Gas Compositional Analysis. The key measures included:

- **Gas Compositional Analysis** (e.g., H<sub>2</sub>, O<sub>2</sub>+Ar, CO<sub>2</sub>, N<sub>2</sub>, CO, C<sub>1</sub>-C<sub>6+</sub>)
- **Gas Isotope Analysis**
  - $\delta^{13}\text{C}_1$  (methane)

- $\delta^{13}\text{C}_2$  (ethane)

The values of methane and ethane isotopes were expressed in parts per thousand (‰) relative to the Vienna Pee Dee Belemnite (VPDB) standard. The goal was to distinguish gas from different reservoir intervals based on isotopic fractionation that can occur during generation and migration.

Each sample's  $\delta^{13}\text{C}_1$  and  $\delta^{13}\text{C}_2$  values were compared to a reference dataset derived from core-based and open-hole gas compositions associated with the Reservoir A and Reservoir B formations. In theory, gases generated from different source rocks or under different thermal regimes exhibit measurable isotopic differences.

However, the overlapping thermal maturity and common source characteristics of Reservoir A and Reservoir B gases in the Field limited the diagnostic resolution of gas isotope data as it will be discussed in more details in the results and discussion section. Therefore, while these measurements helped identify mixed or end-member trends, they were not used in isolation for zonal allocation.

### Water Isotope Analysis & Compositional Analysis

Produced water samples were analyzed for:

- **Major ion composition** (12 (e.g.,  $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Cl}^-$ ,  $\text{SO}_4^{2-}$ ))
- **Strontium isotopes** ( $^{87}\text{Sr}/^{86}\text{Sr}$  ratio)

Water ions analysis was performed to understand the composition of the water prior to performing isotopes analysis which then later was used to support and get better clarity of the water isotopes signature by plotting the TDS vs the  $\text{Na}^+$  composition of each well. Strontium isotope analysis was performed for the selected wells as the  $^{87}\text{Sr}/^{86}\text{Sr}$  ratio is a powerful geochemical tracer because it reflects the signature of the formation rock with which water has equilibrated over geologic time. Unlike major ions, which may be influenced by mixing, evaporation, or scaling, strontium isotopes are considered conservative tracers.

Based on prior core leach studies and limited historical data, the Reservoir A and Reservoir B formations were known to exhibit distinct  $^{87}\text{Sr}/^{86}\text{Sr}$  signatures. Reservoir A formation water typically shows a more radiogenic Sr isotope ratio compared to Reservoir B, likely due to differences in lithology and diagenetic history.

Water samples were plotted on a "Sr isotope vs. Sr (ppm)" which helped to map the wells on the Sr map and draw the end points for each reservoir while wells with mixed signatures plotted along a linear mixing trend, which was interpreted as evidence of commingled production as it will be discussed in more details in the results section.

### Condensate Fingerprinting

Condensate samples were analyzed for extended hydrocarbon fraction composition using gas chromatography. Key molecular markers, including C5–C10 range n-alkanes, were used to define chemical fingerprints unique to each reservoir. Similar to gas and water isotopes, the analysis and results were used along hierarchical clustering to categorize the samples into fluid families and try to map them to different reservoir while comparing the data to PLT results for calibration and to have better understanding of each well.

Condensate geochemistry showed stronger reservoir-level separation than gas isotopes, particularly in wells with limited gas-liquid equilibrium shifts. Integration with water chemistry allowed for multi-phase validation of fluid origin.

## Data Integration and Validation

To test the effectiveness of isotopic fingerprinting, the analytical results were integrated with historical PLT data from selected wells. Crossplots, spider diagrams, and zonal profile maps were constructed to overlay isotopic fingerprints with PLT-determined inflow zones.

Where isotopic endmembers matched PLT interpretations, the data were considered mutually supportive. In cases where discrepancies existed—such as water or condensate signatures inconsistent with PLT results or a response that is very difficult to interpret or understand—additional sampling and data re-evaluation were planned for the next campaign of geochemical fingerprinting.

In several wells, isotope data clarified ambiguous PLT interpretations or revealed dynamic changes in flow profiles that would not have been detected with a one-time PLT survey. These insights formed the basis for improved zonal allocation and reservoir modeling efforts. In general, the water and hydrocarbon isotopes analysis gave the operator a better understanding of the reservoir zonal contributions and reinforced the PLT results in most of the wells and gave different insights about other wells.

## Results and Discussion

This section presents the findings from isotopic and chemical analyses of gas, water, and condensate samples collected across the Field. The results are interpreted in the context of reservoir zonation, production allocation, and integration with historical PLT data.

### Gas Isotope Analysis

The  $\delta^{13}\text{C}$  values for methane (C1) and ethane (C2) across the sampled wells showed relatively narrow variability. Most methane samples ranged between  $-39\text{‰}$  to  $-42\text{‰}$ , while ethane ranged between  $-30\text{‰}$  to  $-33\text{‰}$ . These values fall within the expected range for thermogenic gas generated from marine source rocks at high maturity levels, consistent with both the Reservoir A and Reservoir B formations.

However, the isotopic overlap between the formations limited the utility of gas isotope data for definitive zonal identification. This was anticipated due to the common depositional and thermal history of the two reservoirs. Only a few wells exhibited slightly enriched or depleted  $\delta^{13}\text{C}$  values that could be interpreted as indicative of preferential reservoir contribution. For instance, some wells displayed slightly more enriched  $\delta^{13}\text{C}_1$  values, potentially suggesting a higher Reservoir A contribution. However, these differences were within analytical uncertainty and not considered diagnostic on their own, [Figure, 1](#).

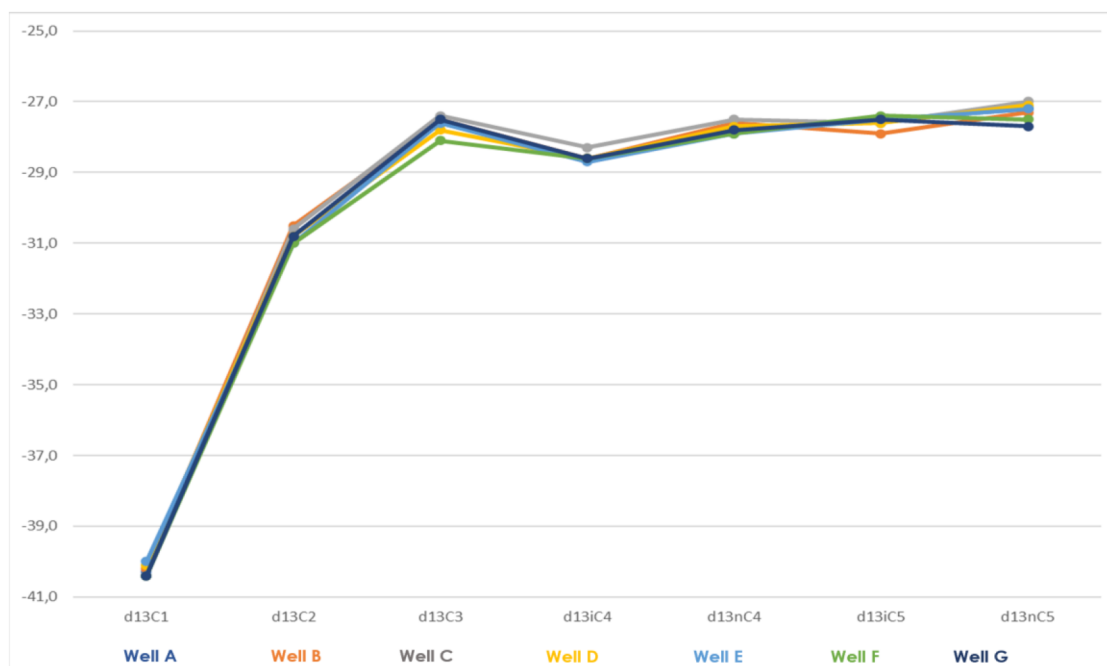


Figure 1—Gas Isotopes showed minimal diagnostic resolution and were not used for zonal allocation.

While gas isotope data did not offer high-resolution differentiation between formations, they were useful for detecting mixed or anomalous production. Wells that plotted between reference endmembers in  $\delta^{13}\text{C}_1$  vs  $\delta^{13}\text{C}_2$  crossplots were interpreted as commingled producers. These findings, when combined with water and condensate fingerprints, added interpretive value. For the operator, the gas isotopes fingerprints were not used for zonal allocation confirmation and the PLT results will remain the main source for gas allocation, however, the recommendation of enhancing gas samples collection and exploring the options of taking more downhole samples when possible, will be considered for future gas fingerprinting campaigns.

### Water Isotope and Geochemical Fingerprinting

Strontium isotope ratios in produced water proved to be a much more reliable discriminator between reservoir zones. The  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios ranged from 0.712 to 0.717, with distinct clustering observed between wells producing predominantly from Reservoir A or Reservoir B formations. This clustering could be summarized as below:

- **Reservoir A dominated wells**, such as well A and well B, exhibited higher, more radiogenic  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios (above 0.715).
- **Reservoir B dominated wells**, including well C and Well D, showed lower ratios (0.7125-0.7135), consistent with the expected Reservoir B isotopic signature.
- **Outlier wells**, including well R that was hydraulically fracked in reservoir B while it was only perforated in reservoir A

These three clustering could be seen when the  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios is plotted against Strontium concentration in ppm, it mapped wells to different reservoirs which agreed with most of the wells to the PLT data and the petrophysical understanding of the water in each well. Figure, 2.



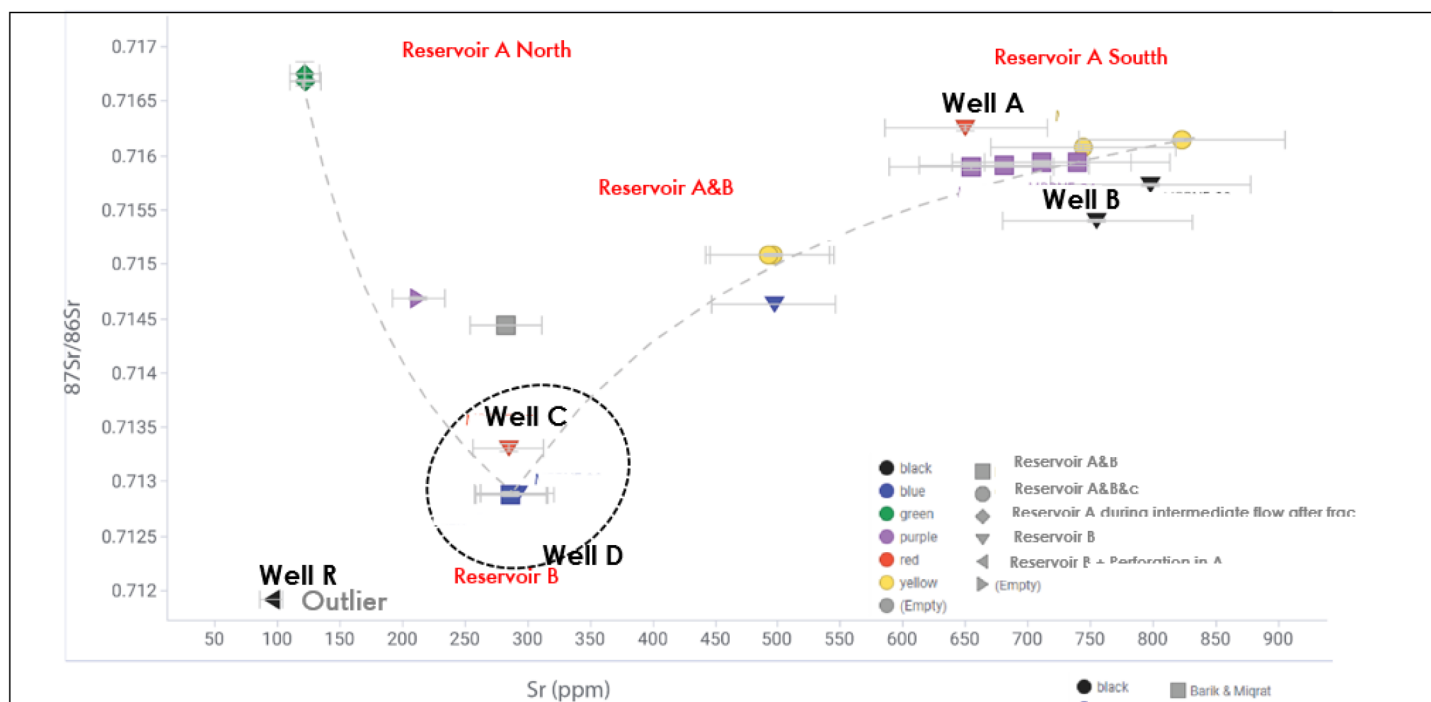


Figure 2— $^{87}\text{Sr}/^{86}\text{Sr}$  ratios Vs. Strontium concentration in ppm.

To have more confidence in the results obtained from Strontium isotopes and concentrations, the Sodium ( $\text{Na}^+$ ) ratios were plotted against the total dissolved solids (TDS) as shown in Figure 3. Mainly two distinct water families emerged, agreeing with what Figure 2 is showing. Moreover, a linear mixing trend was also evident in wells with commingled production.

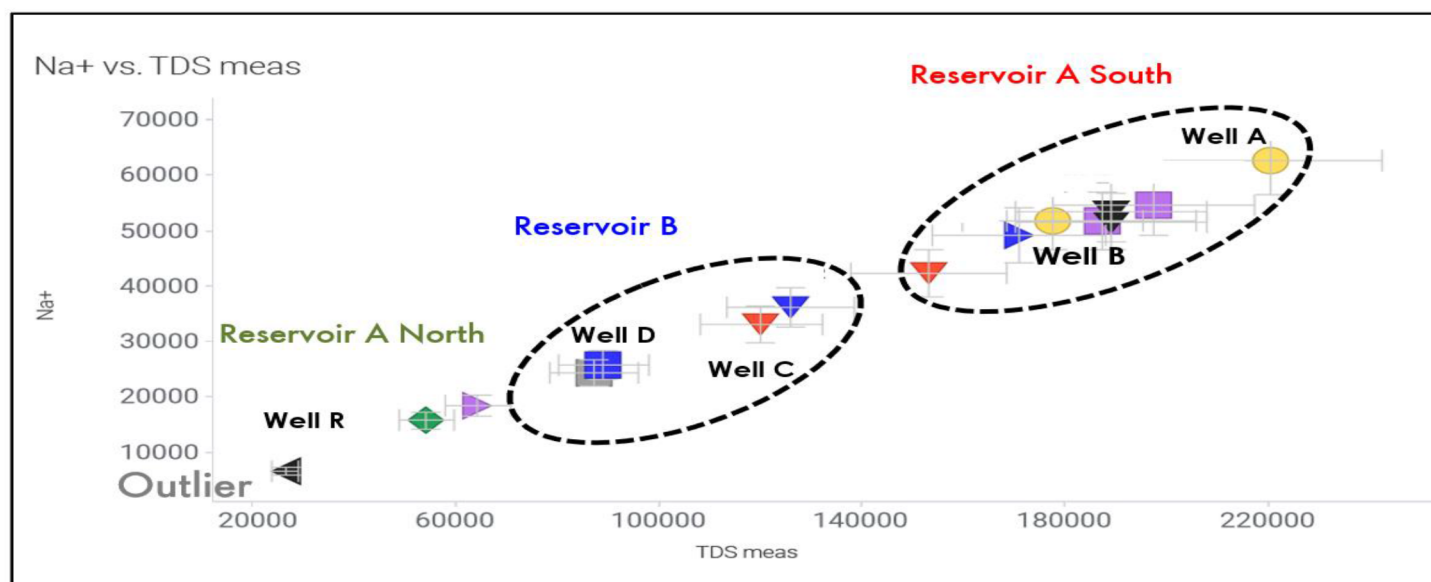


Figure 3—Sodium ( $\text{Na}^+$ ) Vs. Total Dissolved Solids (TDS) plot for the water samples. Wells A, B, C, D, and R are shown as examples.

Similarly, one notable outlier was Well R, which plotted away from both established end members. This suggested either an unidentified water source, water sample contamination, or diagenetic alteration. Additional sampling was recommended to investigate this anomaly.

Overall, the water isotope results provided clear and stable tracers for zonal attribution, with minimal influence from surface separation processes or production phase changes. These results contributed to a deeper understanding of high-water production and reservoir behavior, enhancing decision-making processes and reservoir management strategies

### Condensate Fingerprinting

Condensate composition analysis showed relatively meaningful differentiation between Reservoir A and Reservoir B reservoirs based on molecular ratios and n-alkane profiles as shown in Figure 4. The samples data were grouped into two dominant clusters corresponding with the expected reservoir contributions. End member samples showed distinct groups that helped to establish the baselines for each reservoir, while the comingled wells fell in between these two end members lines

- **Reservoir A-derived condensates** were characterized by higher C7/C6 ratios and an increased abundance of branched alkanes.
- **Reservoir B-derived samples** exhibited longer-chain dominance and a more linear hydrocarbon profile.

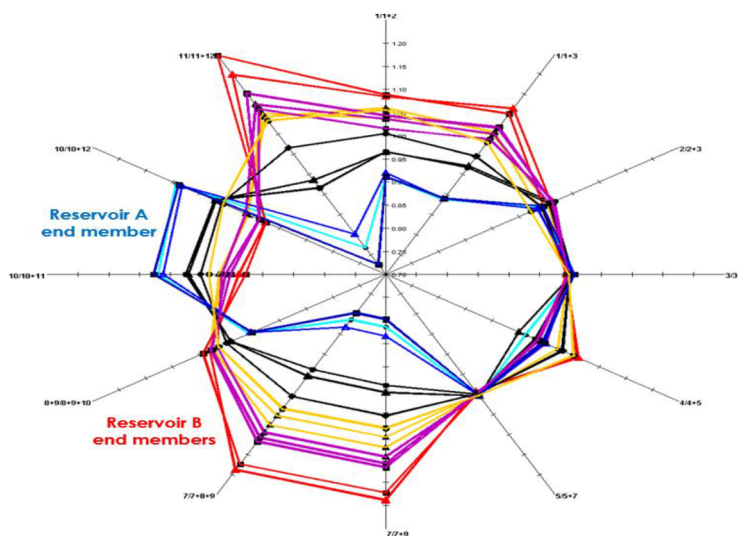


Figure 4—Two dominant clusters corresponding with the expected reservoir condensate contributions.

While the condensate fingerprint showed more meaningful distinction between the samples and helped to confirm reservoir's condensate contributions, it was recommended to enhance the sampling procedure for the condensate as well and to consider taken sampled from wide range of wells in the asset to confirm the results.

### Integration with PLT Data

The integration of isotopic data with historical production logging (PLT) results proved to be a key component of this study. In several cases, isotopic fingerprinting validated the inflow zones previously identified by PLT, thereby increasing confidence in reservoir characterization.

In many instances, isotopic signatures—such as strontium isotopes, water chemistry, and condensate composition—aligned with PLT interpretations, confirming dominant contributions from specific stratigraphic zones.

In other cases, isotopic data offered additional or even contradictory insights. For example, where PLT data was inconclusive due to operational limitations, isotopic analysis revealed fluid sources that deviated from expected profiles. In another scenario, while PLT suggested minimal contribution from a particular



zone, isotopic evidence indicated a more complex inflow pattern, including significant input from that same zone.

This complementary use of isotopic and PLT data highlights the value of multi-disciplinary approaches in refining reservoir understanding and improving zonal attribution.

In all cases, the isotopic data provided either validation or improved understanding, reducing uncertainty around reservoir contributions.

### Operational and Economic Implications

The ability to use fluid fingerprints to infer zonal contributions has direct operational and financial benefits. PLTs, while valuable, are expensive to deploy—particularly in deviated wells or those requiring coil tubing or rig support. Each PLT run in the X Field costs approximately USD 100,000–125,000 and may result in deferred production or well shut-ins.

By reducing the reliance on PLTs for routine surveillance and reserving them for calibration or confirmation, the operator could achieve significant cost savings over the lifetime of the field. The isotope program, while involving analytical expenses, required no well interventions and offered scalable field-wide coverage.

### Conclusions, Recommendations, Novelty and Impact

This study demonstrates the value of integrating isotopic fingerprinting techniques with conventional surveillance tools for enhanced reservoir characterization in complex, commingled production environments. In X Field, Oman—where multiple stacked reservoirs such as the Reservoir A and Reservoir B formations contribute to well production—traditional Production Logging Tools (PLTs) alone are considered insufficient for continuous, cost-effective monitoring of zonal contributions.

The application of strontium isotope analysis in formation water, coupled with hydrocarbon condensate fingerprinting and  $\delta^{13}\text{C}$  gas isotope analysis, has provided an effective method for identifying fluid sources and reservoir contributions. Among the isotopic tracers evaluated,  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios in produced water offered the highest resolution in differentiating between the Reservoir A and Reservoir B reservoirs. Condensate composition analysis further supported reservoir attribution, particularly in cases where gas isotopes showed limited variability.

Integration with historical PLT data confirmed the robustness of isotopic methods. In several wells, isotopic fingerprints validated PLT interpretations, increasing confidence in zonal allocation models. In others, discrepancies between PLT and isotopic data highlighted potential changes in reservoir behavior, previously undetectable through conventional surveillance. Importantly, the isotopic program revealed reservoir dynamics in wells where PLT data was missing, inconclusive, or outdated.

From an operational standpoint, this methodology presents a cost-effective alternative to frequent PLT deployment. The non-intrusive nature of fluid sampling, combined with the scalability of lab-based analysis, enables field-wide surveillance with minimal disruption to production. This is particularly advantageous in mature assets, where well intervention risks and costs are high.

Based on these findings, the following recommendations are proposed:

- Incorporate isotopic and geochemical fingerprinting into routine reservoir surveillance programs, especially in commingled or multi-zone fields.
- Use PLTs selectively for calibration or to resolve complex cases where isotopic data alone may not suffice.
- Develop a centralized reference library of isotopic endmembers for long-term monitoring and model refinement.
- Strengthening the gas and condensate sampling strategy by incorporating enhanced surface protocols and pursuing downhole sampling opportunities where feasible.

Overall, the integrated approach demonstrated in this study can significantly enhance reservoir understanding, reduce surveillance costs, and improve field development decision-making.

This study represents the first comprehensive application of integrated isotopic fingerprinting—specifically strontium isotope analysis and hydrocarbon fluid geochemistry—in X Field, Oman. While PLTs have historically served as the primary diagnostic tool for reservoir inflow profiling, their high cost and operational limitations have created a need for complementary or alternative technologies. This work introduces a novel, non-intrusive surveillance approach that not only augments PLT results but in many cases enhances them.

The use of  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios in produced water as a reservoir diagnostic tool is particularly impactful. This conservative tracer has proven highly effective at distinguishing fluid origins from the Reservoir A and Reservoir B formations, even in wells with complex completion geometries or ambiguous PLT results. The added integration of condensate geochemistry strengthens the methodology by offering phase-independent validation, increasing the robustness of interpretation.

The novelty lies not only in the application of these techniques in a real-field context but also in their demonstrated scalability, cost-effectiveness, and operational simplicity. Unlike PLTs, which require well shut-ins, hardware mobilization, and risk of intervention-related damage, the isotopic program relies on surface samples that can be collected routinely with minimal disruption.

The impact of this approach extends beyond the X Field. It provides a transferable framework for other mature or complex fields facing similar challenges in production allocation and reservoir monitoring. By shifting surveillance from reactive to predictive, this methodology contributes directly to improved reservoir management, better-informed field development decisions, and long-term asset value optimization.

## References

- Birkle, Peter. "Geochemical Fingerprinting of Produced Water for Enhanced Well Control and Fracturing Efficiency." Paper presented at the International Petroleum Technology Conference, Dhahran, Kingdom of Saudi Arabia, January 2020.
- Huc, A. -Y., Carpentier, B., Guéhenneux, G., and C. Magnier. "Geochemistry in a Reservoir and Production Perspective." Paper presented at the Middle East Oil Show and Conference, Bahrain, February 1999.
- Nengkoda, Ardian, Mandhari, Musallam Sulaiman, Widodo, Setiawan, and Zaher Mohammed Al-Hinai. "Evaluation of Reservoir for Production Allocation using Reservoir Fingerprinting: A Case Study." Paper presented at the International Petroleum Technology Conference, Dubai, U.A.E., December 2007.
- Nengkoda, A., Widodo, S., Mandhari, M. S., Hinai, Z., and T. Khan. "The Effectiveness of Geochemical Technique for Evaluation of Commingled Reservoir: Case Study." Paper presented at the Asia Pacific Oil and Gas Conference and Exhibition, Jakarta, Indonesia, October 2007.
- Liu, Faye, Wu, Jiang, Jin, Muqing, Hardman, Douglas L., and Dave Cannon. "From Reservoir Characterization to Reservoir Monitoring: An Integrated Workflow to Optimize Field Development Using Geochemical Fingerprinting Technology." Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Virtual, July 2020.